**DEPARTMENT OF MATHEMATICS**
**UNIVERSITY OF MARYLAND**
**GRADUATE WRITTEN EXAMINATION**
**January, 2021**

**Probability (Ph.D. Version)**

*Instructions to the Student*

a. Answer all six questions. Each will be graded from 0 to 10.

b. Use a different booklet for each question. Write the problem number and your code number (**NOT YOUR NAME**) on the outside cover.

c. Keep scratch work on separate pages in the same booklet.

d. If you use a "well known" theorem in your solution to any problem, it is your responsibility to make clear which theorem you are using and to justify its use.

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**Problem 1.** A frog and a toad jump between the corners of a square. At the end of each minute, the frog either jumps in place (with probability 1/3) or jumps to a neighboring corner (with probability 1/3 each). The toad's jumping patters is slightly different: it always seeks the company of the frog. If the toad is in a corner by itself, it jumps to a neighboring corner with probability 1/2 at the end of the minute. If it is sharing a corner with the frog, it stays in place at the end of the minute. Let $p_n$ be the probability that the frog and the toad are sitting in the same corner at the end of the $n$-th minute. Find the following limit: $\lim_{n\to\infty} p_n$.

**Problem 2.** We will say that a random variable $\xi$ is nontrivial if it is not equal to a constant almost surely, i.e., there does not exist a constant $k$ such that $\mathrm{P}(\xi = k) = 1$.

(a) Does there exist a probability space $(\Omega, \mathcal{F}, \mathrm{P})$ with the sample space that has only three elements (i.e., $\Omega = \{\omega_1, \omega_2, \omega_3\}$) such that there are two nontrivial independent random variables defined on it?

(b) Does there exist a probability space with a finite sample space such that there is an infinite sequence of nontrivial independent identically distributed random variables defined on it?

(c) Does there exist a probability space with a finite sample space such that there is an infinite sequence of nontrivial independent random variables defined on it?

**Problem 3.** Suppose that $\xi_1, \xi_2, ...$ are random variables on $(\Omega, \mathcal{F}, \mathrm{P})$ such that $\mathrm{Var}(\xi_n) \leq 1$ for each $n$ and $\mathrm{Cov}(\xi_m, \xi_n) \leq 0$ for each $m \neq n$. Prove that the Weak Law of Large Numbers holds for the sequence $S_n = \xi_1 + ... + \xi_n$ (i.e., $(S_n - \mathrm{E}S_n)/n \to 0$ in probability).

**Problem 4.** Let $\xi_1, \xi_2, ...,$ defined on $(\Omega, \mathcal{F}, \mathrm{P})$ be independent identically distributed Bernoulli random variables with $\mathrm{P}(\xi_n = 1) = p \in (0, 1)$. Let $A_n$ be the event that $n$ of these random variables in a row, starting from an index between $2^n$ and $2^{n+1}$, are equal to 1, i.e.,

$$A_n = \{\text{there is } m \in [2^n, 2^{n+1}] \text{ such that } \xi_m = \xi_{m+1} = ... = \xi_{m+n-1} = 1\}.$$

(a) Prove that almost every $\omega$ belongs on only finitely many of these events if $p < 1/2$.
(b) Prove that almost every $\omega$ belongs to infinitely many of these events if $p \geq 1/2$.

**Problem 5.** Let $W_t^1$ and $W_t^2$ be independent Wiener processes (Brownian motions). Find $\mathrm{P}(\text{there exists } t \in [0, 1] \text{ such that } W_t^1 \geq W_t^2 + 1)$.

**Problem 6.** Let $\xi_1, \xi_2, ...$ be independent identically distributed random variables such that

$$\mathrm{P}(\xi_n = 1) = p \in (0, 1), \quad \mathrm{P}(\xi_n = -1) = q = 1 - p.$$

Let $\mathcal{F}_n = \sigma(\xi_1, ..., \xi_n)$ be the natural filtration and $S_n = \xi_1 + ... + \xi_n$.
(a) Let $X_n = (q/p)^{S_n}$. Prove that $(X_n, \mathcal{F}_n)$, $n \geq 1$, is a martingale.
(b) Find $c > 0$ such that $(c^n 2^{S_n}, \mathcal{F}_n)$ is a martingale.